

SOLICITATION/CONTRACT/ORDER FOR COMMERCIAL ITEMS OFFEROR TO COMPLETE BLOCKS 12, 17, 23, 24, & 30				1. REQUISITION NUMBER NB305000-26-00183		PAGE 1 OF 22					
2. CONTRACT NO.		3. AWARD/EFFECTIVE DATE		4. ORDER NUMBER		5. SOLICITATION NUMBER 1333ND26QNB030152		6. SOLICITATION ISSUE DATE			
7. FOR SOLICITATION INFORMATION CALL:		a. NAME ERIK FRYCKLUND EDF@MAIL.NIST.GOV				b. TELEPHONE NUMBER (No collect calls)		8. OFFER DUE DATE/ LOCAL TIME MAY 12, 2026 1:00 PM ET			
		9. ISSUED BY NATIONAL INST OF STDS AND TECHNOLOGY 100 BUREAU DRIVE STOP 1640 BUILDING 301 ROOM B129 GAITHERSBURG MD 20899-1640				10. THIS ACQUISITION IS ☐ UNRESTRICTED OR ☒ SET ASIDE: <u>100</u> % FOR: ☒ SMALL BUSINESS ☐ WOMEN-OWNED SMALL BUSINESS (WOSB) ELIGIBLE UNDER THE WOMEN-OWNED SMALL BUSINESS PROGRAM NAICS: 334413 ☐ HUBZONE SMALL BUSINESS ☐ EDWOSB ☐ SERVICE-DISABLED VETERAN-OWNED SMALL BUSINESS ☐ 8 (A) SIZE STANDARD: 1250					
11. DELIVERY FOR FOB DESTINATION UNLESS BLOCK IS MARKED ☐ SEE SCHEDULE		12. DISCOUNT TERMS		13a. THIS CONTRACT IS A RATED ORDER UNDER DPAS (15 CFR 700) ☐		13b. RATING					
						14. METHOD OF SOLICITATION ☒ RFQ ☐ IFB ☐ RFP					
15. DELIVER TO NATIONAL INST OF STDS AND TECHNOLOGY 100 BUREAU DRIVE STOP 1640 BUILDING 301 ROOM B129 GAITHERSBURG MD 20899-1640		CODE 000SB		16. ADMINISTERED BY CODE							
17a. CONTRACTOR/ OFFEROR. CODE		FACILITY CODE		18a. PAYMENT WILL BE MADE BY CODE							
TELEPHONE NO.		17b. CHECK IF REMITTANCE IS DIFFERENT AND PUT SUCH ADDRESS IN OFFER ☐		18b. SUBMIT INVOICES TO ADDRESS SHOWN IN BLOCK 18a UNLESS BLOCK BELOW IS CHECKED ☐ SEE ADDENDUM							
19. ITEM NO.		20. SCHEDULE OF SUPPLIES/SERVICES			21. QUANTITY		22. UNIT	23. UNIT PRICE		24. AMOUNT	
Please		see continuation page for line item details.									
		(Use Reverse and/or Attach Additional Sheets as Necessary)									
25. ACCOUNTING AND APPROPRIATION DATA See Schedule							26. TOTAL AWARD AMOUNT (For Govt. Use Only)				
☐ 27a. SOLICITATION INCORPORATES BY REFERENCE FAR 52.212-1, 52.212-4. FAR 52.212-3 AND 52.212-5 ARE ATTACHED. ADDENDA							☒ ARE ☐ ARE NOT ATTACHED				
☐ 27b. CONTRACT/PURCHASE ORDER INCORPORATES BY REFERENCE FAR 52.212-4. FAR 52.212-5 IS ATTACHED. ADDENDA							☐ ARE ☐ ARE NOT ATTACHED				
☒ 28. CONTRACTOR IS REQUIRED TO SIGN THIS DOCUMENT AND RETURN COPIES TO ISSUING OFFICE. CONTRACTOR AGREES TO FURNISH AND DELIVER ALL ITEMS SET FORTH OR OTHERWISE IDENTIFIED ABOVE AND ON ANY ADDITIONAL SHEETS SUBJECT TO THE TERMS AND CONDITIONS SPECIFIED					1		☐ 29. AWARD OF CONTRACT: REF. _____ OFFER DATED _____. YOUR OFFER ON SOLICITATION (BLOCK 5), INCLUDING ANY ADDITIONS OR CHANGES WHICH ARE SET FORTH HEREIN, IS ACCEPTED AS TO ITEMS:				
30a. SIGNATURE OF OFFEROR/CONTRACTOR					31a. UNITED STATES OF AMERICA (SIGNATURE OF CONTRACTING OFFICER)						
30b. NAME AND TITLE OF SIGNER (Type or print)			30c. DATE SIGNED		31b. NAME OF CONTRACTING OFFICER (Type or print)			31c. DATE SIGNED			

19. ITEM NO.	20. SCHEDULE OF SUPPLIES/SERVICES	21. QUANTITY	22. UNIT	23. UNIT PRICE	24. AMOUNT

32a. QUANTITY IN COLUMN 21 HAS BEEN

☐ RECEIVED
 ☐ INSPECTED
 ☐ ACCEPTED, AND CONFORMS TO THE CONTRACT, EXCEPT AS NOTED: _____

32b. SIGNATURE OF AUTHORIZED GOVERNMENT REPRESENTATIVE	32c. DATE	32d. PRINTED NAME AND TITLE OF AUTHORIZED GOVERNMENT REPRESENTATIVE
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32e. MAILING ADDRESS OF AUTHORIZED GOVERNMENT REPRESENTATIVE	32f. TELEPHONE NUMBER OF AUTHORIZED GOVERNMENT REPRESENTATIVE
	32g. E-MAIL OF AUTHORIZED GOVERNMENT REPRESENTATIVE

33. SHIP NUMBER	34. VOUCHER NUMBER	35. AMOUNT VERIFIED CORRECT FOR	36. PAYMENT ☐ COMPLETE ☐ PARTIAL ☐ FINAL	37. CHECK NUMBER
☐ PARTIAL ☐ FINAL				

38. S/R ACCOUNT NO.	39. S/R VOUCHER NUMBER	40. PAID BY
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41a. I CERTIFY THIS ACCOUNT IS CORRECT AND PROPER FOR PAYMENT	42a. RECEIVED BY (Print)
41b. SIGNATURE AND TITLE OF CERTIFYING OFFICER	42b. RECEIVED AT (Location)
41c. DATE	42c. DATE REC'D (YY/MM/DD)
	42d. TOTAL CONTAINERS

SOLICITATION

Procurement of Silicon and Silicon/Germanium Epitaxial Wafers

This is a solicitation for commercial items prepared in accordance with the format in FAR Subpart 12.201–Solicitation Procedures for Commercial Items, as supplemented with additional information included in this notice. This announcement constitutes the only solicitation; quotations are being requested, and a separate written solicitation document will not be issued. This solicitation is being issued under the authority of FAR Part 12 Acquisition of Commercial Products.

This solicitation is a Request for Quotations (RFQ). The solicitation document and incorporated provisions and clauses are those in effect through Department of Commerce FAC Deviation effective April 3, 2026.

The associated North American Industrial Classification System (NAICS) code for this procurement shall be 334413–Semiconductor and Related Device Manufacturing with a small business size standard of 1250 employees.

The National Institute of Standards and Technology (NIST), CHIPS Metrology program develops and advances cutting edge metrology capabilities for members of the US semiconductor manufacturing ecosystem. This NIST conducted research program works with device manufacturers, tool vendors, materials suppliers, and other organizations to address critical metrology gaps to spur innovation within seven grand challenge areas. CHIPS Metrology researchers at NIST require high-quality epitaxial film stacks on silicon wafers. Although we anticipate changes in the exact specifications of each wafer lot purchased, the general classes of wafer products can be well described.

Responsible quoters shall provide pricing for the following Line Items:

LINE ITEMS:

- LINE ITEM 0001 – “Thin” Wafers (Quantity 15), in accordance with the attached document.
- LINE ITEM 0002 – “Thin” Wafers (Quantity 15), in accordance with the attached document.
- LINE ITEM 0003 – Optional lines item “Thin” Wafers (Quantity Min 10), in accordance with the attached document.
- LINE ITEM 0004 – Optional lines item “Thin” Wafers (Quantity Min 10), in accordance with the attached document.
- LINE ITEM 0005 – Optional lines item “Thin” Wafers (Quantity Min 10), in accordance with the attached document.
- LINE ITEM 0006 – Optional lines item “Thin” Wafers (Quantity Min 10), in accordance with the attached document.
- LINE ITEM 0007 – Optional lines item “Thin” Wafers (Quantity Min 10), in accordance with the attached document.

DELIVERY

Delivery: Shall occur within 20 weeks of award of this requirement.

Questions regarding this Solicitation can be emailed to erik.frycklund@nist.gov

EVALUATION CRITERIA AND BASIS FOR AWARD

The Government intends to award a purchase order resulting from this solicitation to the responsible quoter whose

quotation, conforming to the solicitation, is the lowest price, technically acceptable quotation.

Technically acceptable means that the Contractor's provide a products meeting the minimum requirements in the requirements document.

Price will be evaluated for all quoters whose quotations are determined technically acceptable. Price will be evaluated to determine that total price is consistent with the technical portion of the quotation and is a fair and reasonable overall price to the Government.

The Contracting Officer may, if necessary, request revised quotation(s) prior to award.

REQUIRED SUBMISSIONS

All Contractors shall submit the following:

1. For the purpose of evaluation of Technical Acceptability:

Quoters shall provide documentation that demonstrates the quoted product's ability to meet or exceed the minimum specifications in accordance with the requirements document.

For the purpose of Price:

Quoters shall provide firm-fixed price quotations for Line Items

2. Quoters shall include a completed copy of all required solicitation provisions contained in the attached Applicable Provisions/Clauses document (required provisions are highlighted)
3. Any award resulting from this solicitation will contain the following statement: "The Government's terms and conditions, identified herein, shall be the prevailing terms and conditions governing this award. In the event of a conflict between the Government's and contractor's terms and conditions, the Government's terms and conditions take precedence." If the Contractor objects to any of the terms and conditions contained in this solicitation, the contractor shall state "The terms and conditions in the solicitation are acceptable to be included in the award document with the exception, deletion, or addition of the following:" [Contractor shall list exception(s) and rationale for the exception(s)]. It is the sole responsibility of the Contractor to identify in their quote any exceptions to the terms and conditions of the solicitation. If the contractor does not include such a statement, the submission of a quotation in response to this solicitation will be regarded as the Contractor's acceptance of the Government's terms and conditions for inclusion into the resultant purchase order as prevailing.
 4. Quoters shall provide an active Government Unique Identifier number for the System for Award Management (SAM) registration. Quoters must have an active registration at www.SAM.Gov in order to be technically evaluated.
 5. Quoters shall provide the location of product manufacturer to be either domestic or foreign. If non domestic, the country of origin must be listed.

Please note that this procurement IS NOT being conducted under the GSA Federal Supply Schedule (FSS) program or another Government-Wide Area Contract (GWAC). If a contractor submits a quotation based upon an FSS or GWAC contract, the Government will accept the quoted price. However, the terms and conditions stated herein will be included in any resultant Purchase Order, not the terms and conditions of the contractor's FSS or GWAC contract, and the statement required above shall be included in the quotation.

PROVISIONS AND CLAUSES:

The clauses and provisions applicable to this solicitation can be found under Applicable Clauses & Provisions. The full text of a FAR provision or clause may be accessed electronically at www.acquisition.gov/far. The full text of a CAR provision or clause may be accessed electronically at <https://www.acquisition.gov/far-overhaul/far-part-deviation-guide>

ATTACHMENTS:

The following attachments apply to this solicitation:

1. The Requirements Document
2. Applicable Clauses & Provisions

SCHEDULE Continued

ITEM NO.	SUPPLIES/SERVICES	QUANTITY	UNIT	UNIT PRICE	AMOUNT
0001	Wafers in accordance with the attached requirements document. Thin Si and SiGe layers on Si - Spec 1	15.00	EA		
0002	Thin Si and SiGe layers on Si - Spec 2	15.00	EA		
0003	Thin doped Si on Si	10.00	EA		
0004	Thick doped Si on Si	10.00	EA		
0005	Isotopically enriched 28Si on Si	10.00	EA		
0006	Thin Si and SiGe layers on Si	10.00	EA		
0007	Thick SiGe on buffered Si	10.00	EA		

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CLAUSES

Requirement

STATEMENT OF REQUIREMENT

Title: Silicon and Silicon/Germanium Epitaxial Wafers

I. BACKGROUND INFORMATION

The CHIPS Metrology program develops and advances cutting edge metrology capabilities for members of the US semiconductor manufacturing ecosystem. This NIST conducted research program works with device manufacturers, tool vendors, materials suppliers, and other organizations to address critical metrology gaps to spur innovation within seven grand challenge areas. For more information on CHIPS Metrology, please visit <https://www.nist.gov/chips/research-development-programs/metrology-program>

Silicon wafers with epitaxial layer of silicon and silicon/germanium are an essential item for modern semiconductor manufacturing. Some manufacturers perform epitaxial growths in house while other film stacks can be purchased. Wafer with specific compositions, thicknesses, and dopant levels, and supporting measurements are available at various wafer sizes and lot sizes. NIST's requirement is for small lot sizes (small number of wafers) and small wafers (100 mm diameter, not the more common commercial size of 300 mm diameter).

II. PURPOSE

CHIPS Metrology researchers at NIST require high-quality epitaxial film stacks on silicon wafers. Although we anticipate changes in the exact specifications of each wafer lot purchased, the general classes of wafer products can be well described.

The anticipated contract includes base line items with full wafer specifications. Additionally, there are optional line items where the exact specifications for lot size, layer count, thicknesses, and layer composition/doping will be specified upon ordering. Additionally, Secondary Ion Mass Spectroscopy (SIMS) and Spreading Resistance Profiling (SRP) data for each lot of wafers must be available for purchase as well.

III. MINIMUM REQUIREMENTS

Line Item 0001:

Description: "Thin" Si and SiGe layers on Si – Spec 1

Quantity: 15 wafers

1. Substrate
 - 1.1. Prime double side polished (DSP) silicon
 - 1.2. Orientation: <100> with SEMI standard primary wafer flat
 - 1.3. Thickness: 525 $\mu\text{m} \pm 25 \mu\text{m}$
 - 1.4. Dopant: phosphorus, resistivity range 1 $\Omega\text{-cm}$ to 100 $\Omega\text{-cm}$
2. Film Stack per table below

Layer	Thickness [nm]	Composition [at]	Dopant [$\text{at}\cdot\text{cm}^{-3}$]
7	10 ± 2	Si	B, 2×10^{14} to 5×10^{15}
6	10 ± 2	$\text{Si}_{0.7}\text{Ge}_{0.3}$	B, 5×10^{19} to 5×10^{20}
5	10 ± 2	Si	B, 2×10^{14} to 5×10^{15}
4	10 ± 2	$\text{Si}_{0.7}\text{Ge}_{0.3}$	B, 5×10^{19} to 5×10^{20}
3	10 ± 2	Si	B, 2×10^{14} to 5×10^{15}
2	10 ± 2	$\text{Si}_{0.7}\text{Ge}_{0.3}$	B, 5×10^{19} to 5×10^{20}
1	10 ± 2	Si	B, 2×10^{14} to 5×10^{15}
Substrate			

Line Item 0002:

Description: "Thin" Si and SiGe layers on Si – Spec 2

Quantity: 15 wafers

1. Substrate
 - 1.1. Prime double side polished (DSP) silicon
 - 1.2. Orientation: <100> with SEMI standard primary wafer flat
 - 1.3. Thickness: 525 $\mu\text{m} \pm 25 \mu\text{m}$
 - 1.4. Dopant: phosphorus, resistivity range 1 $\Omega\text{-cm}$ to 100 $\Omega\text{-cm}$

2. Film Stack per table below

Layer	Thickness [nm]	Composition [at frac]	Dopant [at:cm ⁻³]
7	25 ± 5	Si	B, 2×10 ¹⁴ to 5×10 ¹⁵
6	20 ± 5	Si _{0.55} Ge _{0.45}	
5	25 ± 5	Si	
4	30 ± 5	Si _{0.70} Ge _{0.30}	
3	25 ± 5	Si	
2	40 ± 5	Si _{0.85} Ge _{0.15}	
1	25 ± 5	Si	
Substrate			

Line Item 0003: OPTION LINE ITEM 1

Description: “Thin” doped Si on Si

Quantity: At least 10 wafers, lot size specified at time of order

1. Fixed Parameters
 - 1.1. Substrate: 100 mm prime double side polished <100> Si wafers, 525 µm thick, SEMI standard flats.
 - 1.2. Film stack: Fully coherent Si with B and Ph dopants at concentrations up to 5×10²⁰ cm⁻³ (atomic, no requirement for electrically active). Total film stack thickness up to 1 µm with up to 5 layers, with no layers thinner than 10 nm.
 - 1.3. Minimum lot size: 10 wafers
2. Parameters configurable at time of Modification
 - 2.1. Substrate resistivity: The dopant (B, N, Ph, As, and Sb) and target resistivity (within ±10×)
 - 2.2. Film stack: The thickness and doping of each layer
 - 2.3. Lot size: Minimum order quantity if 10 wafers, but additional wafers of the same lot may be ordered with an additional per wafer cost.

Line Item 0004: OPTION LINE ITEM 2

Description: “Thick” doped Si on Si

Quantity: At least 10 wafers, lot size specified at time of Modification

1. Substrate: 100 mm prime double side polished <100> Si wafers, 525 µm thick, SEMI standard flats.
 - 1.1. Film stack: Fully coherent Si with B and Ph dopants at concentrations up to 5×10²⁰ cm⁻³. Total film thickness up to 10 µm in one layer.
 - 1.2. Minimum lot size: 10 wafers
2. Parameters configurable at time of order
 - 2.1. Substrate resistivity: The dopant (B, N, Ph, As, and Sb) and target resistivity (within ±10×)
 - 2.2. Film stack: The thickness and doping of the doped Si layer
 - 2.3. Lot size: Minimum order quantity if 10 wafers, but additional wafers of the same lot may be ordered with an additional per wafer cost.

Line Item 0005: OPTION LINE ITEM 3

Description: Isotopically enriched ²⁸Si on Si

Quantity: At least 10 wafers, lot size specified at time of Modification

1. Fixed Parameters
 - 1.1. Substrate: 100 mm prime double side polished <100> Si wafers, 525 µm thick, SEMI standard flats.
 - 1.2. Film stack: Fully coherent Si or ²⁸Si with B, N, and Ph dopants at concentrations up to 5×10²⁰ cm⁻³. Total film thickness up to 100 nm with no layers thinner than 5 nm.
 - 1.3. Minimum lot size: 10 wafers
2. Parameters configurable at time of order
 - 2.1. Substrate resistivity: The dopant (B, N, Ph, As, and Sb) and target resistivity (within ±10×)
 - 2.2. Film stack: The thickness, doping, and isotopic composition of each layer.

Line Item 0006: OPTION LINE ITEM 4

Description: “Thin” Si and SiGe layers on Si

Quantity: At least 10 wafers, lot size specified at time of Modification

1. Fixed Parameters

- 1.1. Substrate: 100 mm prime double side polished <100> Si wafers, 525 μm thick, SEMI standard flats.
- 1.2. Film stack: Fully coherent Si and $\text{Si}_{1-x}\text{Ge}_x$ superlattices where x is between 0.1 and 0.4. Total film thickness up to 250 nm with no layer thinner than 5 nm. B and Ph dopants at concentrations up to $5 \times 10^{20} \text{ cm}^{-3}$.
- 1.3. Minimum lot size: 10 wafers
2. Parameters configurable at time of order
 - 2.1. Substrate resistivity: The dopant (B, N, Ph, As, or Sb) and target resistivity (within $\pm 10\times$)
 - 2.2. Film stack: The thickness, Ge composition, and doping of each layer
 - 2.3. Lot size: Minimum order quantity if 10 wafers, but additional wafers of the same lot may be ordered with an additional per wafer cost.

Line Item 0007: OPTION LINE ITEM 5

Description: "Thick" SiGe on buffered Si

Quantity: At least 10 wafers, lot size specified at time of Modification

1. Fixed Parameters
 - 1.1. Substrate: 100 mm prime double side polished <100> Si wafers, 525 μm thick, SEMI standard flats.
 - 1.2. Film stack: Virtual substrate created by relaxed buffer layer(s) of SiGe with a fully relaxed (relative to Si) $\text{Si}_{1-x}\text{Ge}_x$ layer on top where x is between 0.1 and 0.4. Thickness of the relaxed SiGe film thickness less than 10 μm .
 - 1.3. Minimum lot size: 10 wafers
2. Parameters configurable at time of order
 - 2.1. Substrate resistivity: The dopant (B, N, Ph, As, or Sb) and target resistivity (within $\pm 10\times$)
 - 2.2. Film stack: Relaxed SiGe thickness, Ge composition, and doping of the relaxed SiGe layer. Vendors may create a virtual substrate with layers of SiGe buffers that matches the needs of the specified relaxed SiGe layer.
 - 2.3. Lot size: Minimum order quantity if 10 wafers, but additional wafers of the same lot may be ordered with an additional per wafer cost.

IV. SCHEDULE OF DELIVERABLES

Define all deliverables and indicate (1) the quantity desired; (2) the format in which each deliverable is to be supplied; (3) the delivery date (based on the number of weeks/days); (4) the place of delivery. The context of each deliverable shall be clear.

<i>Deliverable Number</i>	<i>Description</i>	<i>Quantity</i>	<i>Due Date</i>	<i>Place of Delivery</i>
1	CLIN 001 CLIN 002	15 wafers 15 wafers	Within 20 weeks of award	NIST Gaithersburg
2	Remaining CLINs		Within 20 weeks of option modification	NIST Gaithersburg

Option Line items shall be valid for three years after acceptance of CLINs 0001 and 0002.

V. DELIVERY TERMS

Delivery shall be F.O.B Destination and shall occur in accordance with the delivery due dates provided in the below table.

FOB Destination means: The contractor shall pack and mark the shipment in conformance with carrier requirements, deliver the shipment in good order and condition to the point of delivery specified in the purchase order, be responsible for any loss of and/or damage to the goods occurring before receipt and acceptance of the shipment by the consignee at the delivery point specified in the purchase order; and pay all charges to the specified point of delivery.

The contractor shall deliver all Line Items to:
National Institute of Standards and Technology
100 Bureau Drive, Building 301
Gaithersburg, MD 20899

Delivery shall take place during normal business hours, between 7:30 am and 3:00 pm Eastern Time, Monday through Friday except Federal Holidays, and shall be coordinated with the NIST Subject Matter Expert.

VI. INSPECTION & ACCEPTANCE

In addition to the inspection and acceptance terms articulated in the specific FAR clause that allows the Government reserves the right to perform such performance tests and evaluations as defined below to verify specified system performance. Such tests and evaluations, if performed, shall be conducted within the environment that the system is to be operated. The Contractor has the right to be present during the tests and evaluations, if performed, at the Contractor's expense.

Performance Tests – as appropriate for each class of wafer product:

1. X-ray Reflectivity (XRR) for film thickness
2. High-resolution X-ray Diffraction (HR-XRD) for coherent SiGe compositions
3. Secondary Ion Mass Spectroscopy (SIMS) for dopant concentrations (data may be supplied by vendor if optional SIMS data is included in purchase)
4. Spreading Resistance Profiling (SRP) for active dopant quantification

Ownership of the equipment shall transfer to NIST upon acceptance by the Government.

The Government will test, inspect, and accept or reject the equipment within 14 of the receipt of the items unless otherwise indicated above.

PROVISIONS AND CLAUSES

52.252-1 Solicitation Provisions Incorporated by Reference.

SOLICITATION PROVISIONS INCORPORATED BY REFERENCE (FEB 1998)

This solicitation incorporates one or more solicitation provisions by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. The offeror is cautioned that the listed provisions may include blocks that must be completed by the offeror and submitted with its quotation or offer. In lieu of submitting the full text of those provisions, the offeror may identify the provision by paragraph identifier and provide the appropriate information with its quotation or offer. Also, the full text of a solicitation provision may be accessed electronically at this/these address(es):

FAR Overhaul (DEVIATIONS JAN 2026): <https://www.acquisition.gov/far-overhaul/far-part-deviation-guide>

CAR: <https://www.ecfr.gov/current/title-48/chapter-13>

(End of provision)

PROVISIONS INCORPORATED BY REFERENCE:

52.203-18 Prohibition on Contracting with Entities that Require Certain Internal Confidentiality Agreements or Statements-Representation. (DEVIATION JAN 2026)

52.204-7 System For Award Management (DEVIATION JAN 2026)

52.209-2 Prohibition on Contracting with Inverted Domestic Corporations-Representation (DEVIATION Jan 2026)

52.212-1 Instructions To Offerors--Commercial Products And Commercial Services (DEVIATION JAN 2026)

52.222-18 Certification Regarding Knowledge of Child Labor for Listed End Products (DEVIATION JAN 2026)

52.252-2 CLAUSES INCORPORATED BY REFERENCE (FEB 1998)

This contract incorporates one or more clauses by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. Also, the full text of a clause may be accessed electronically at this/these address(es):

FAR Overhaul (DEVIATIONS JAN 2026): <https://www.acquisition.gov/far-overhaul/far-part-deviation-guide>

CAR: <https://www.ecfr.gov/current/title-48/chapter-13>

(End of clause)

CLAUSES INCORPORATED BY REFERENCE:

52.203-17 Contractor Employee Whistleblower Rights (Nov 2023)

52.203-19, Prohibition on Requiring Certain Internal Confidentiality Agreements or Statements (Jan 2017)

52.204-13 System For Award Management Maintenance (Deviation Jan 2026)

52.204-19 Incorporation By Reference Of Representations And Certifications (Dec 2014)

52.209-6 Protecting the Government's Interest When Subcontracting With Contractors Debarred, Suspended, Proposed for Debarment, or Voluntarily Excluded (DEVIATION JAN 2026)

52.209-10, Prohibition on Contracting with Inverted Domestic Corporations (DEVIATION JAN 2026)

52.219-6, Notice of Total Small Business Set-Aside (Nov 2020)

52.219-28, Post-award Small Business Program Rerepresentation (DEVIATION JAN 2026)
 52.222-3, Convict Labor (Jun 2003)
 52.222-19, Child Labor—Cooperation with Authorities and Remedies (DEVIATION JAN 2026)
 52.222-36, Equal Opportunity for Workers with Disabilities (DEVIATION JAN 2026)
 52.222-50, Combating Trafficking in Persons (DEVIATION JAN 2026)
 52.222-90 Addressing DEI Discrimination by Federal Contractors (DEVIATION JAN 2026)
 52.223-23, Sustainable Products and Services (May 2024)
 52.225-1, Buy American-Supplies (DEVIATION JAN 2026)
 52.226-8, Encouraging Contractor Policies to Ban Text Messaging While Driving (May 2024)
 52.232-29 Terms for Financing of Commercial Products and Commercial Services (Nov 2021)
 52.232-33, Payment by Electronic Funds Transfer-System for Award Management (Oct2018)
 52.232-39 Unenforceability of Unauthorized Obligations (JUNE 2013)
 52.232-40, Providing Accelerated Payments to Small Business Subcontractors (Mar 2023)
 52.233-3, Protest After Award (DEVIATION JAN 2026)
 52.233-4, Applicable Law for Breach of Contract Claim (Oct 2004)
 52.244-6 Subcontracts for Commercial Products and Commercial Services (DEVIATION JAN 2026)

1352.201-70 Contracting Officer Authority (APR 2010)
 1352.209-73 Compliance with The Laws (APR 2010)
 1352.209-74 Organizational Conflict of Interest (APR 2010)

Full Text Provision and Clauses:

FAR 52.212-4 TERMS AND CONDITIONS—COMMERCIAL PRODUCTS AND COMMERCIAL SERVICES (JANUARY 2026)

- (a) Definitions. The clause at Federal Acquisition Regulation (FAR) 52.202-1, Definitions, is incorporated by reference.
- (b) Inspection/Acceptance. The Contractor shall only tender for acceptance those items that conform to the requirements of this contract. The Government reserves the right to inspect or test any supplies or services that have been tendered for acceptance. The Government may require repair or replacement of nonconforming supplies or reperformance of nonconforming services at no increase in contract price. If repair/replacement or reperformance will not correct the defects or is not possible, the Government may seek an equitable price reduction or adequate consideration for acceptance of nonconforming supplies or services. The Government must exercise its post acceptance rights—
- (1) Within a reasonable time after the defect was discovered or should have been discovered; and
 - (2) Before any substantial change occurs in the condition of the item, unless the change is due to the defect in the item.
- (c) Assignment. The Contractor or its assignee may assign its rights to receive payment due as a result of performance of this contract to a bank, trust company, or other financing institution, including any Federal lending agency in accordance with the Assignment of Claims Act (31 U.S.C. 3727). However, when a third party makes payment (e.g., use of the Governmentwide commercial purchase card), the Contractor may not assign its rights to receive payment under this contract.
- (d) Changes. Changes in the terms and conditions of this contract may be made only by written agreement of the parties.
- (e) Disputes. This contract is subject to 41 U.S.C. chapter 71, Contract Disputes. Failure of the parties to this contract to reach agreement on any request for equitable adjustment, claim, appeal, or action arising under or relating to this contract shall be a dispute to be resolved in accordance with the clause FAR 52.233-1,

Disputes, which is incorporated in this contract by reference. The Contractor shall proceed diligently with performance of this contract, pending final resolution of any dispute arising under the contract.

(f) Excusable delays. The Contractor shall be liable for default unless nonperformance is caused by an occurrence beyond the reasonable control of the Contractor and without its fault or negligence. Examples of occurrences include acts of God or the public enemy, acts of the Government in either its sovereign or contractual capacity, fires, floods, epidemics, quarantine restrictions, strikes, unusually severe weather, and delays of common carriers. When an excusable delay occurs, the Contractor shall—

- (1) Notify the Contracting Officer in writing as soon as possible;
- (2) Remedy the delay as quickly as possible; and
- (3) Notify the Contracting Officer when the occurrence is over.

(g) Invoice. The Government will handle invoices according to the Prompt Payment Act (31 U.S.C. 3903) and 5 CFR part 1315. The Contractor shall submit invoices to the address designated in the contract to receive invoices. An invoice must include the information required by 5 CFR part 1315.9(b).

(h) Patent indemnity. The Contractor shall indemnify the Government and its officers, employees, and agents against liability, including costs, for actual or alleged direct or contributory infringement of, or inducement to infringe, any United States or foreign patent, trademark, or copyright, arising out of the performance of this contract, provided the Contractor is reasonably notified of such claims and proceedings.

(i) Payment—

(1) Items accepted. Payment shall be made for items accepted by the Government that have been delivered to the delivery destinations set forth in this contract.

(2) Prompt payment. The Government will make payment in accordance with the Prompt Payment Act (31 U.S.C. 3903) and prompt payment regulations at 5 CFR part 1315.

(3) Discount. In connection with any discount offered for early payment, time shall be computed from the date of the invoice. For the purpose of computing the discount earned, payment shall be considered to have been made on the date that appears on the payment check or the specified payment date if an electronic funds transfer payment is made.

(4) Overpayments. If the Contractor becomes aware of a duplicate contract financing or invoice payment or that the Government has otherwise overpaid on a contract financing or invoice payment, the Contractor shall—

(i) Remit the overpayment amount to the payment office cited in the contract along with a description of the overpayment including the—

(A) Circumstances of the overpayment (e.g., duplicate payment, erroneous payment, liquidation errors, date(s) of overpayment);

(B) Affected contract number and delivery order number, if applicable;

(C) Affected line item or subline item, if applicable;

(D) Contractor point of contact; and

(ii) Provide a copy of the remittance and supporting documentation to the Contracting Officer.

(5) Interest.(i) All amounts that become payable by the Contractor to the Government under this contract shall bear simple interest from the date due until paid unless paid within 30 days of becoming due. The interest rate shall be the interest rate established by the Secretary of the Treasury as provided in 41 U.S.C. 7109, which is applicable to the period in which the amount becomes due, as provided in (i)(6)(v) of this

clause, and then at the rate applicable for each six-month period as fixed by the Secretary until the amount is paid.

(ii) The Government may issue a demand for payment to the Contractor upon finding a debt is due under the contract.

(iii) Final decisions. The Contracting Officer will issue a final decision as required by FAR part 33 if—

(A) The Contracting Officer and the Contractor are unable to reach agreement on the existence or amount of a debt within 30 days;

(B) The Contractor fails to liquidate a debt previously demanded by the Contracting Officer within the timeline specified in the demand for payment unless the amounts were not repaid because the Contractor has requested an installment payment agreement; or

(C) The Contractor requests a deferment of collection on a debt previously demanded by the Contracting Officer (see FAR part 32).

(iv) If a demand for payment was previously issued for the debt, the demand for payment included in the final decision shall identify the same due date as the original demand for payment.

(v) Amounts shall be due at the earliest of the following dates:

(A) The date fixed under this contract.

(B) The date of the first written demand for payment, including any demand for payment resulting from a termination for cause.

(vi) The interest charge shall be computed for the actual number of calendar days involved beginning on the due date and ending on—

(A) The date on which the designated office receives payment from the Contractor;

(B) The date of issuance of a Government check to the Contractor from which an amount otherwise payable has been withheld as a credit against the contract debt; or

(C) The date on which an amount withheld and applied to the contract debt would otherwise have become payable to the Contractor.

(vii) The interest charge made under this clause may be reduced under the procedures for interest credits prescribed in FAR part 32 in effect on the date of this contract.

(j) Risk of loss. Unless the contract specifically provides otherwise, risk of loss or damage to the supplies provided under this contract shall remain with the Contractor until, and shall pass to the Government upon—

(1) Delivery of the supplies to a carrier, if transportation is f.o.b. origin; or

(2) Delivery of the supplies to the Government at the destination specified in the contract, if transportation is f.o.b. destination.

(k) Taxes. The contract price includes all applicable Federal, State, and local taxes and duties.

(l) Termination for the Government's convenience. The Government reserves the right to terminate this contract, or any part hereof, for its sole convenience. In the event of such termination, the Contractor shall immediately stop all work and shall immediately cause any and all of its suppliers and subcontractors to cease work. Subject to the terms of this contract, the Contractor shall be paid a percentage of the contract price reflecting the percentage of the work performed prior to the notice of termination, plus reasonable charges the Contractor can demonstrate to the satisfaction of the Government using its standard record keeping system, have resulted from the termination. The Contractor shall not be required to comply with the cost

accounting standards or contract cost principles for this purpose. This paragraph does not give the Government any right to audit the Contractor's records. The Contractor shall not be paid for any work performed or costs incurred which reasonably could have been avoided.

(m) Termination for cause. The Government may terminate this contract, or any part hereof, for cause in the event of any default by the Contractor, or if the Contractor fails to comply with any contract terms and conditions, or fails to provide the Government, upon request, with adequate assurances of future performance. The Government will send a cure notice to the Contractor, unless the reason for the termination is late delivery. In the event of termination for cause, the Government shall not be liable to the Contractor for any amount for supplies or services not accepted, and the Contractor shall be liable to the Government for any and all rights and remedies provided by law. If it is determined that the Government improperly terminated this contract for default, such termination shall be deemed a termination for convenience.

(n) Title. Unless specified elsewhere in this contract, title to items furnished under this contract shall pass to the Government upon acceptance, regardless of when or where the Government takes physical possession.

(o) Warranty. The Contractor warrants and implies that the items delivered under this contract are merchantable and fit for use for the particular purpose described in this contract.

(p) Limitation of liability. Except as otherwise provided by an express warranty, the Contractor will not be liable to the Government for consequential damages resulting from any defect or deficiencies in accepted items.

(q) Compliance with laws unique to Government contracts. The Contractor agrees to comply with 31 U.S.C. 1352 relating to limitations on the use of appropriated funds to influence certain Federal contracts; 40 U.S.C. chapter 37, Contract Work Hours and Safety Standards; 41 U.S.C. chapter 87, Kickbacks; 49 U.S.C. 40118, Government-financed air transportation; and 41 U.S.C. chapter 21 relating to procurement integrity.

(r) Order of precedence. Any inconsistencies in this solicitation or contract shall be resolved by giving precedence in the following order:

- (1) The schedule of supplies/services;
 - (2) The Disputes, Payments, Invoice, Compliance with Laws Unique to Government Contracts, and Unauthorized Obligations paragraphs of this clause;
 - (3) Other contract clauses incorporated in the solicitation or contract;
 - (4) Addenda to this solicitation or contract;
 - (5) Solicitation provisions incorporated in the solicitation;
 - (6) Other paragraphs of this clause;
 - (7) Other documents, exhibits, and attachments; and
 - (8) The specification.
- (s) Unauthorized obligations.

(1) Except as stated in paragraph (s)(2) of this clause, when any supply or service acquired under this contract is subject to any End User License Agreement (EULA), Terms of Service (TOS), or similar legal instrument or agreement, that includes any clause requiring the Government to indemnify the Contractor or any person or entity for damages, costs, fees, or any other loss or liability that would create an Anti-Deficiency Act violation (31 U.S.C. 1341), the following shall govern:

- (i) Any such clause is unenforceable against the Government.
- (ii) Neither the Government nor any Government-authorized end user shall be deemed to have agreed to

such clause by virtue of it appearing in the EULA, TOS, or similar legal instrument or agreement. If the EULA, TOS, or similar legal instrument or agreement is invoked through an "I agree" click box or other comparable mechanism (e.g., "click-wrap" or "browse-wrap" agreements), execution does not bind the Government or any Government authorized end user to such clause.

(iii) Any such clause is deemed to be stricken from the EULA, TOS, or similar legal instrument or agreement.

(2) Paragraph (s)(1) of this clause does not apply to indemnification by the Government that is expressly authorized by statute and specifically authorized under applicable agency regulations and procedures.

(t) Comptroller General examination of record. This paragraph applies if this contract was awarded using other than sealed bid procedures and is in excess of the simplified acquisition threshold on the date of award of this contract.

(1) The Comptroller General of the United States, or an authorized representative of the Comptroller General, shall have access to and right to examine any of the Contractor's directly pertinent records involving transactions related to this contract.

(2) The Contractor shall make available at its offices, at all reasonable times, the records, materials, and other evidence for examination, audit, or reproduction, until 3 years after final payment under this contract or for any shorter period specified in FAR part 4, longer period required by statute, or periods specified in other clauses of this contract. If this contract is completely or partially terminated, the records relating to the work terminated shall be made available for 3 years after any resulting final termination settlement. Records relating to appeals under the disputes clause or to litigation or the settlement of claims arising under or relating to this contract shall be made available until such appeals, litigation, or claims are finally resolved.

(3) As used in this clause, records include books, documents, accounting procedures and practices, and other data, regardless of type and regardless of form. This clause does not require the Contractor to create or maintain any record that the Contractor does not maintain in the ordinary course of business or pursuant to a provision of law.

(u) Incorporation by reference. The Contractor's representations and certifications, including those completed electronically via the System for Award Management (SAM), are incorporated by reference into the contract.

(End of clause)

52.217-7 Option for Increased Quantity-Separately Priced Line Item. (Mar 1989)

The Government may require the delivery of the numbered line item, identified in the Schedule as an option item, in the quantity and at the price stated in the Schedule. The Contracting Officer may exercise the option by written notice to the Contractor within *30 days*. Delivery of added items shall continue at the same rate that like items are called for under the contract, unless the parties otherwise agree.

(End of clause)

52.217-9 Option to Extend the Term of the Contract. (Mar 2000)

(a) The Government may extend the term of this contract by written notice to the Contractor within 30 days provided that the Government gives the Contractor a preliminary written notice of its intent to extend at least 30 days before the contract expires. The preliminary notice does not commit the Government to an extension.

(b) If the Government exercises this option, the extended contract shall be considered to include this option clause.

(c) The total duration of this contract, including the exercise of any options under this clause, shall not exceed 3 years.

(End of clause)

52.225-2 BUY AMERICAN CERTIFICATE (DEVIATION JAN 2026)

(a)

(1) The Offeror certifies that each end product, except those listed in paragraph (b) of this provision, is a domestic end product and that each domestic end product listed in paragraph (c) of this provision contains a critical component.

(2) The Offeror shall list as foreign end products those end products manufactured in the United States that do not qualify as domestic end products. For those foreign end products that do not consist wholly or predominantly of iron or steel or a combination of both, the Offeror shall also indicate whether these foreign end products exceed 55 percent domestic content, except for those that are COTS items. If the percentage of the domestic content is unknown, select "no".

(3) The Offeror shall separately list the line item numbers of domestic end products that contain a critical component (see FAR 25.105).

(4) The terms "commercially available off-the-shelf (COTS) item," "critical component," "domestic end product," "end product," and "foreign end product" are defined in the clause of this solicitation entitled "Buy American-Supplies."

(b) Foreign End Products:

[illegible]

(b) *List as necessary*

(c) Domestic end products containing a critical component:

Line Item No. _____

[List as necessary]

(d) The Government will evaluate offers in accordance with the policies and procedures of part 25 of the Federal Acquisition Regulation.

(End of provision)

52.225-18 PLACE OF MANUFACTURE (DEVIATION JAN 2026)

(a) Definitions. As used in this provision—

Manufactured end product means any end product in product and service codes (PSCs) 1000-9999, except-

(1) PSC 5510, Lumber and Related Basic Wood Materials;

(2) Product or Service Group (PSG) 87, Agricultural Supplies;

(3) PSG 88, Live Animals:

(4) PSG 89, Subsistence;

(5) PSC 9410, Crude Grades of Plant Materials;

(6) PSC 9430, Miscellaneous Crude Animal Products, Inedible;

(7) PSC 9440, Miscellaneous Crude Agricultural and Forestry Products;

(8) PSC 9610, Ores:

- (9) PSC 9620, Minerals, Natural and Synthetic; and
- (10) PSC 9630, Additive Metal Materials.

Place of manufacture means the place where an end product is assembled out of components, or otherwise made or processed from raw materials into the finished product that is to be provided to the Government. If a product is disassembled and reassembled, the place of reassembly is not the place of manufacture.

(b) For statistical purposes only, the offeror shall indicate whether the place of manufacture of the end products it expects to provide in response to this solicitation is predominantly-

- (1) ☒ In the United States (Check this box if the total anticipated price of offered end products manufactured in the United States exceeds the total anticipated price of offered end products manufactured outside the United States); or
- (2) ☐ Outside the United States.

(End of provision)

1352.233-70 AGENCY PROTESTS (APR 2010)

(a) An agency protest may be filed with either: (1) the contracting officer, or (2) at a level above the contracting officer, with the appropriate agency Protest Decision Authority. See 64 Fed. Reg. 16,651 (April 6, 1999)

(b) Agency protests filed with the Contracting Officer shall be sent to the following address:

ATTN: Donald Graham CONTRACTING OFFICER
100 Bureau Drive, MS 1640
Gaithersburg, MD 20899

(c) Agency protests filed with the agency Protest Decision Authority shall be sent to the following address: ATTN: NIST PROTEST CONTROLLER

100 Bureau Drive, MS 1640
Gaithersburg, MD 20899

(d) A complete copy of all agency protests, including all attachments, shall be served upon the Contract Law Division of the Office of the General Counsel within one day of filing a protest with either the Contracting Officer or the Protest Decision Authority.

(e) Service upon the Contract Law Division shall be made as follows:

U.S. Department of Commerce
Office of the General Counsel
Chief, Contract Law Division
Room 5893
Herbert C. Hoover Building
14th Street and Constitution Avenue, N.W.
Washington, D.C. 20230.
FAX: (202) 482-5858

(End of clause)

1352.233-71 GAO AND COURT OF FEDERAL CLAIMS PROTESTS (APR 2010)

(a) A protest may be filed with either the Government Accountability Office (GAO) or the Court of Federal Claims unless an agency protest has been filed.

(b) A complete copy of all GAO or Court of Federal Claims protests, including all attachments, shall be served upon (i) the Contracting Officer, and (ii) the Contract Law Division of the Office of the General Counsel, within one day of filing a protest with either GAO or the Court of Federal Claims.

(c) Service upon the Contract Law Division shall be made as follows:

U.S. Department of Commerce
Office of the General Counsel
Chief, Contract Law Division

Room 5893
Herbert C. Hoover Building
14th Street and Constitution Avenue Washington, D.C. 20230.
FAX: (202) 482-5858
(End of clause)

1352.246-70 PLACE OF ACCEPTANCE (APR 2010)

(a) The Contracting Officer or the duly authorized representative will accept supplies and services to be provided under this contract.

(b) The place of acceptance will be:
National Institute of Standards and Technology
100 Bureau Drive,
Gaithersburg, MD 20899
(End of clause)

NIST LOCAL-53 CONTRACT PERFORMANCE DURING CHANGES IN NIST OPERATING STATUS

All contractors performing work on active contracts at the U.S. Department of Commerce (DOC), National Institute of Standards and Technology (NIST) campuses and/or working in NIST workspaces should go to the www.nist.gov website and under the "About NIST" tab click on "Visit". This site includes information about campus access and security information; identification requirements; parking information and more.

Contractor personnel are required to check the appropriate campus operating status and personnel requirements at <https://www.nist.gov/campus-status> daily prior to arriving on site. All personnel must adhere to the requirements set forth in the operating status.

Unless otherwise stated in the contract terms and conditions, normal days of business operation are Monday through Friday, excluding Federal Holidays. However, throughout the contract period of performance, there may be circumstances beyond the control of NIST that will impact normal days of business operation such as inclement weather, power outages, etc. In circumstances such as these, the Contractor must call the appropriate NIST campus status line to verify the operating status:

Gaithersburg Campus Operating Status Line:
(301) 975-8000
(800) 437-4385 x8000 (toll free)
Boulder Campus Operating Status Line:
(303) 497-4000
(303) 497-3000 option 2

In the event of a lapse in appropriation, access to Government facilities and resources, including equipment and systems will be limited to excepted personnel for both Federal employees and contractor personnel. If performance of the contract is onsite and/or requires Government interaction, unless the contractor has been, or is notified that it is required to work under an excepted status, the contractor must stop work. The work stoppage shall remain in effect until the lapse is resolved and notification is provided via the NIST website at www.nist.gov (banner on front page) and/or the NIST operating status line(s). Additionally, contractors are encouraged to monitor public broadcasts or the Office of Personnel Management's website at www.opm.gov for the Federal Government operating status.

NIST will provide notification to all contractors that are determined to have excepted status. All excepted contractors are required to continue performance and communicate with the appointed Contracting Officer's Representative (COR) for further guidance, or NIST Contracting Officer if a COR is not appointed.

Contractors with active supply or service contracts that are fully funded at the time of contract award and do not require access to Government facilities, resources, or active administration by

Government personnel in a manner that would not cause the Government to incur additional obligations during the lapse in appropriation may continue performance.

Please note that in all circumstances that impact operations on the NIST campuses, contractors are expected to follow all direction guidance provided by NIST authorities.

(End of clause)

NIST LOCAL-54 ELECTRONIC BILLING INSTRUCTIONS

NIST requires that Invoice/Voucher submissions are sent electronically via email to INVOICE@NIST.GOV.

Each Invoice or Voucher submitted shall include the following:

- (1) Contract number;
- (2) Contractor name and address;
- (3) Unique entity identifier (see www.sam.gov for the designated entity for establishing unique entity identifiers);
- (4) Date of invoice;
- (5) Invoice number;
- (6) Amount of invoice and cumulative amount invoiced to-date;
- (7) Contract Line Item Number (CLIN);
- (8) Description, quantity, unit of measure, unit price, and extended price of supplies/services delivered;
- (9) Prompt payment discount terms, if offered; and
- (10) Any other information or documentation required by the contract.

(End of clause)

NIST LOCAL-56 INVOICING PROCESSING PLATFORM-ALTERNATE I (DEC 2022)

Upon written notice from the contracting officer the following supersedes all other instructions for the submission of payment requests. Accordingly, following written notice payment requests must be submitted electronically through the U.S. Department of the Treasury's Invoice Processing Platform System (IPP).

"Payment request" means any request for contract financing payment or invoice payment by the Contractor. To constitute a proper invoice, the payment request must comply with the requirements identified in the applicable payment request or invoicing instructions, Prompt Payment clause included in the contract, or the clause 52.212-4 Contract Terms and Conditions - Commercial Items included in commercial item contracts. The IPP website address is <https://www.ipp.gov>.

Under this contract, the following documents are required to be submitted as an attachment to the IPP invoice:

The Contractor must use the IPP website to register, access, and use IPP for submitting payment requests. If not already enrolled, the Contractor Government Business Point of Contact (as listed in SAM) will receive enrollment instructions via email within three to five business days of the addition of the contract award to IPP. Contractor assistance with enrollment can be obtained by contacting the IPP Production Helpdesk via email: IPPCustomerSupport@fiscal.treasury.gov or phone (866) 973-3131.

If the Contractor is unable to comply with the requirement to use IPP for submitting payment requests, the Contractor must submit a waiver request in writing to the Contracting Officer with its proposal or quotation. Contact the contracting officer for more information on submitting a waiver request

(End of clause)

